



CSE-476: Introduction to Natural Language Processing
Quiz 3

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In a standard RNN, what is the primary purpose of the "hidden state"?
- (A) To store the final output of the entire sequence.
 - (B) To act as a memory that carries information from previous time steps
 - (C) To calculate the loss function for the current word
 - (D) To normalize the input embeddings
- Q2. Why is the Softmax function typically used in the final layer of a Neural Language Model?
- (A) To ensure the output values sum to 1, representing a probability distribution
 - (B) To prevent the gradients from exploding
 - (C) To increase the speed of the training process
 - (D) To allow the model to handle variable-length sequences
- Q3. What is a key limitation of feed-forward neural networks for NLP tasks?
- (A) They cannot use embeddings
 - (B) They assume fixed input size and struggle with temporal order
 - (C) They cannot be trained with gradient descent
 - (D) They do not use activation functions
- Q4. What is the purpose of the "Forward Pass" in training?
- (A) To update weights based on the loss
 - (B) To calculate the predicted output and compute the loss
 - (C) To unroll the network into a sequence
 - (D) To calculate the gradient for each parameter
- Q5. What is true about the weights U , V , and W across different timestamps in an RNN?
- (A) They are randomly re-initialized at every step
 - (B) They are reused across timestamps

- (C) They increase in value as the sequence progresses
 (D) Only the output weight V is reused
- Q6. Suppose all activation functions in a 3-layer neural network are linear. What is the effective model equivalent to?
- (A) A deeper network with more representational power
 (B) A single linear model
 (C) A recurrent neural network
 (D) A decision tree
- Q7. When training an RNN for language modeling using teacher forcing, the model predicts:
- (A) The next word given the predicted history
 (B) The next word given the ground-truth history
 (C) The current word given itself
 (D) All words simultaneously
- Q8. In an RNN, the hidden state at time t depends on:
- (A) Only the input at time t
 (B) Only the hidden state at time $t - 1$
 (C) Both the input at time t and the hidden state at time $t - 1$
 (D) All future inputs
- Q9. Consider a neural network layer defined as: $h = f(Wx)$. The input vector x has shape $(n_0 \times 1)$, and the output vector h has shape $(n_1 \times 1)$. What must be the shape of W ?
- (A) $(n_0 \times n_1)$
 (B) $(n_1 \times n_0)$
 (C) $(n_0 \times 1)$
 (D) $(1 \times n_1)$
- Q10. What is the main role of backpropagation in neural network training?
- (A) To compute predictions during inference
 (B) To calculate gradients of the loss with respect to parameters
 (C) To initialize weights randomly
 (D) To normalize input data

End of Paper